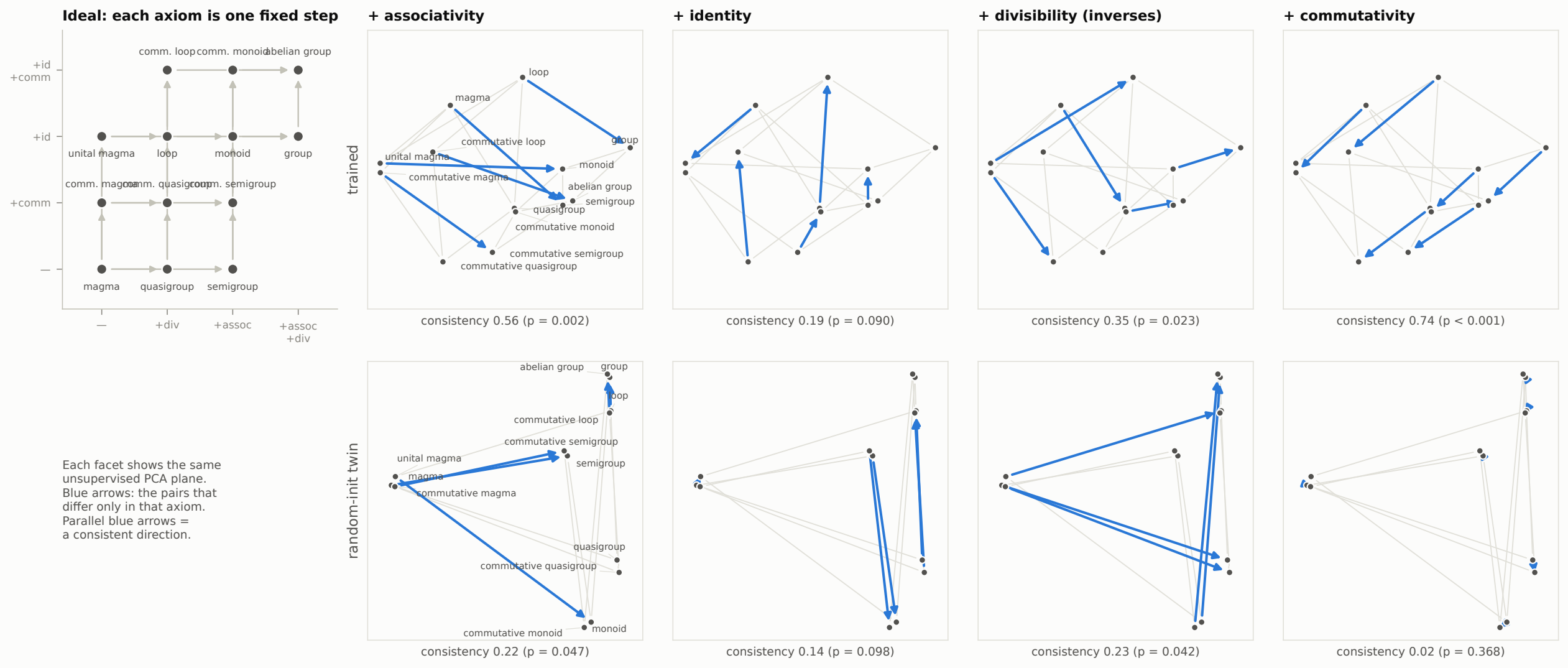
Ideal: each axiom is one fixed step
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
comm. loop
comm. monoid
abelian group
+id +comm
+id
unital magma
loop
monoid
group
comm. magma
comm. quasigroup
comm. semigroup
+comm
magma
quasigroup
semigroup
—
+div
+assoc
+assoc +div
trained
loop
magma
commutative loop
group
unital magma
monoid
commutative magma
abelian group
semigroup
quasigroup
commutative monoid
commutative semigroup
commutative quasigroup
consistency 0.56 (p = 0.002)
consistency 0.19 (p = 0.090)
consistency 0.35 (p = 0.023)
consistency 0.74 (p < 0.001)
random-init twin
abelian group
group
loop
commutative loop
commutative semigroup
unital magma
semigroup
magma
commutative magma
quasigroup
commutative quasigroup
commutative monoid
monoid
consistency 0.22 (p = 0.047)
consistency 0.14 (p = 0.098)
consistency 0.23 (p = 0.042)
consistency 0.02 (p = 0.368)
Each facet shows the same unsupervised PCA plane. Blue arrows: the pairs that differ only in that axiom. Parallel blue arrows = a consistent direction.